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Minimum Cost to Fill Given Weight

Difficulty: **Medium**Accuracy: **25.56%**Submissions: **78K+**Points: **4**

Given a bag of size w kg and you are provided costs of packets different weights of oranges in array $cost[]$, find the minimum total cost to buy exactly w kg oranges

- The cost of 1 kg orange is present at index 0 and in general $arr[i]$ has cost of $(i+1)$ kg orange.
- $cost[i] = -1$ means that ' $i+1$ ' kg packet of orange is unavailable.
- If it is not possible to buy exactly w kg oranges then return -1. It may be assumed that there is an infinite supply of all available packet types.

Examples:

Input: $cost[] = [20, 10, 4, 50, 100]$, $w = 5$ **Output:** 14**Explanation:** The minimum cost is 14 by purchasing a 2kg packet for 10 and a 3kg packet for 4.**Input:** $cost[] = [-1, -1, 4, 3, -1]$, $w = 5$ **Output:** -1**Explanation:** It is not possible to buy 5 kgs.

```
1 class Solution:
2     def minimumCost(self, cost, w):
3         INF = float('inf')
4
5         dp = [INF] * (w + 1)
6         dp[0] = 0
7
8         for i in range(len(cost)):
9             if cost[i] == -1:
10                 continue
11
12             weight = i + 1
13
14             for j in range(weight, w + 1):
15                 dp[j] = min(dp[j], dp[j - weight] + cost[i])
16
17         return -1 if dp[w] == INF else dp[w]
```

Constraints: $1 \leq \text{cost.size()}, w \leq 2 \cdot 10^3$  $\text{cost}[i] \leq 10^5$ [Try more examples](#)**Expected Complexities****Topic Tags**[Custom Input](#)[Compile & Run](#)[Submit](#)